9

Take-home messages

The power of GNN-based recommender systems stems from their ability to directly leverage the user-item interaction graph. However, this power is not monolithic; it is shaped by a complex interplay between model architecture and data characteristics. Throughout this monograph, we have systematically deconstructed these systems by establishing a formal pipeline (Section 3), defining the classical and topological properties of their data (Section 4-6), and empirically measuring their interplay through a rigorous explanatory framework (Section 7). This comprehensive analysis now allows us to distill our findings into a series of actionable take-home messages. We begin by focusing on the GNNs themselves, examining how fundamental architectural choices dictate their performance and behavior. Subsequently, we will delve into the properties of the data, providing guidance on how to study and connect them to a model's expected performance. These messages are designed to bridge the gap between theoretical analysis and practical application, providing clear guidelines for building and evaluating the next generation of GNN-based recommender systems.

9.1 Architectural choices drive performance and behavior

The specific design of a GNN is not a trivial implementation detail; it is the primary determinant of its performance, biases, and limitations within recommender systems. As our model taxonomy (Section 3) and empirical analysis (Section 7) have shown, architectural choices create fundamental trade-offs between complexity, accuracy, and beyond-accuracy objectives. The following messages distill these critical lessons learned from the evolution of GNN architectures:

- **Simplicity often outperforms complexity.** As detailed in our model taxonomy, the evolution from complex early models like NGCF to simpler architectures like LightGCN marked a pivotal moment. Our analysis confirms that removing non-linearities and feature transformations often leads to superior accuracy and efficiency, demonstrating that the core strength of GNNs in this domain is effective neighborhood aggregation, not convoluted transformations (He *et al.*, 2020).
- **Over-smoothing is the primary architectural bottleneck.** The tendency for node embeddings to become indistinguishable after a few message-passing steps is a fundamental limitation discussed in Section 3.3.2. This not only restricts most models to shallow depths but also negatively impacts beyond-accuracy metrics like novelty and diversity (Anelli *et al.*, 2022b). This is why models like SVD-GCN, which we analyzed, avoid iterative message-passing to capture higher-order signals without degradation.
- **Standard GNNs can amplify popularity bias.** The message-passing mechanism inherently favors nodes with high degrees. Our own experimental results in Section 7 (Figure 7.2) support this, showing a consistent positive correlation between the item Gini coefficient (a measure of popularity concentration) and model performance. This confirms that GNNs tend to reward popular items, a critical consideration for fairness (Anelli *et al.*, 2023a).
- **Contrastive learning is a powerful tool for data sparsity.** To address the data sparsity defined in Section 4.3, architectures like

SGL and SimGCL have proven effective. By creating augmented views of the graph or embeddings, these models learn more robust representations that are less dependent on the local neighborhood structure, improving performance in sparse data regimes (Wu *et al.*, 2021; Yu *et al.*, 2022).

9.2 Relationship of classical dataset characteristics with performance

Even though classical dataset characteristics like shape or density (Section 4) have been investigated previously for more simple recommendation algorithms, the relationship between such characteristics and GNN-based recommenders is a novel contribution of our work, as presented in Section 7. The most remarkable insights, as evidenced in Figure 7.2, can be summarized as:

- **GNN models benefit from more (dense) data available.** Consistent positive correlation between density and performance, throughout datasets and GNN models, evidence that when the GNN has access to less sparse information, it can create more meaningful representations of users and items, hence producing more accurate recommendations.

- **Too many users may be detrimental.** By the interplay between space size and shape properties, we observed that the raw number of potential interactions (space size) is not a consistent predictor of performance, whereas the relationship between the number of users and items (shape) is always inversely related to performance. This implies that, for a fixed number of items, increasing the number of users will negatively impact performance; symmetrically, for a given number of users, having more items will produce a lower shape, which has been shown to be related to better performance.

- **Skewed item distributions produce better models.** While this is not true for user distributions (as measured by Gini user), since correlations are either negligible or not significant, for item

distributions (Gini item), the correlation values are very high in every case. This means that GNN models need item distributions where popular and non-popular items are easily discriminated from each other, because of their tendency to promote popular items (hence, if the distributions are too flat, the promoted items may not be the most appropriate ones).

9.3 Topological characterization for performance prediction

While classical dataset characteristics, such as sparsity and density (Section 4), provide a necessary high-level overview, our work demonstrates that they tell only part of the story. To understand the potential and challenges of a dataset for a GNN-based recommender, practitioners must examine the user-item graph's structure in greater detail. The most potent insights come from its topological properties. As we defined and re-interpreted in Section 5, these properties translate abstract graph theory into tangible recommendation concepts:

- **Node degree** is the most direct measure of user activity and item popularity, immediately quantifying the extent of the cold-start problem.
- **Clustering coefficient** captures the strength of the collaborative filtering signal, revealing the density of shared preferences within local neighborhoods. A high clustering coefficient implies a strong foundation for neighborhood-based methods.
- **Degree assortativity** uncovers the homophily within the graph. It tells us whether active users tend to engage with items that are also popular among other active users, a deeper, "look-ahead" property that classical metrics cannot capture.
- **Degree distribution** suggests the scale-free tendency of the user-item graph, where very few nodes have high degree (i.e., active users and popular items), while the majority of nodes present low degrees.

This is not merely a theoretical exercise. The explanatory framework in Section 7 provides strong empirical evidence that these topological

features are powerful and often dominant predictors of GNN performance. Our regression analysis (Figure 7.2) consistently showed that characteristics like average user degree and average user clustering coefficient have a statistically significant and positive impact on accuracy across different models and datasets.

Therefore, a critical take-home message is to make topological analysis a standard, non-negotiable step in any GNN-based recommendation project. This analysis moves beyond a simple sparsity diagnosis to a much richer understanding. It allows for the quantification of the inherent strengths and weaknesses of a dataset, sets realistic performance expectations, and guides the selection of an appropriate GNN architecture before investing in costly training and evaluation cycles.

9.4 Unveiling topological properties within model formalizations

A GNN-based recommender does not learn from graph-structured data if not explicitly defined to do so; its mathematical formulation is explicitly or implicitly designed to embed the topological properties of that data. Understanding this connection is fundamental to moving beyond treating these models as black boxes. Our analysis in Section 6, which dissected the equations of various GNNs, reveals that these properties are encoded in distinct ways, with significant consequences for model behavior.

The most direct and explicit encoding is node degree. As our review of model architectures showed, nearly every modern GNN for recommendation, from NGCF to LightGCN to SGL, incorporates node degree directly into its message-passing formulation via symmetric normalization. The $1/(\sqrt{\sigma_u}\sqrt{\sigma_i})$ term is not merely a mathematical trick for stabilization; it is a deliberate architectural choice that rescales the contribution of messages based on user activity and item popularity. This makes the model's learning process fundamentally dependent on the node degree distribution of the training graph.

Other crucial properties, like the clustering coefficient and degree assortativity, are captured implicitly. These characteristics are not represented by a specific variable in the GNN's equations. Instead, they are an emergent consequence of the multi-hop message-passing mechanism itself.

- **Clustering** is implicitly learned through 2-hop paths (e.g., user-item-user). When two users have a high clustering coefficient, it means they share many common items. The message-passing process naturally aggregates information from these shared items, pulling the embeddings of these two users closer together in the latent space. Thus, the architecture inherently learns the collaborative filtering signal that the clustering coefficient measures.

- **Assortativity**, the tendency for nodes of similar degrees to connect, is captured over even deeper propagation paths. A multi-layer GNN can learn whether active users tend to interact with items that are also popular among other active users. This "look-ahead" property, as we termed it in Section 5, is implicitly encoded as the model learns to navigate longer relationships in the graph.

The take-home message is that a model's architecture determines its topological sensitivity. The explicit reliance on node degree means that if the degree distribution of the training data is not representative, the model may be biased and fail to generalize. Furthermore, the depth of a GNN dictates its ability to implicitly capture more complex properties. A shallow, 1-layer model may only effectively leverage node degree, while a deeper model can begin to harness the signals from clustering and assortativity. This understanding helps explain the performance variations we observed in Section 7 and provides a powerful tool for predicting how a given model will behave on a new dataset with a unique topological signature.

9.5 Connecting data topology to model performance

The ultimate goal of our analysis is to move beyond the costly and inefficient trial-and-error model selection paradigm. A recommender system's performance is not an intrinsic property of the model alone, but an emergent outcome of the dynamic interplay between a model's architectural sensitivities and the dataset's unique topological signature. Understanding and predicting this interplay is the fundamental step toward building more targeted and effective GNN-based systems.

Our explanatory framework, empirically validated in Section 7, provides the tools to forge this connection. By first studying the topological properties of the data and then understanding how different GNNs embed these properties, we can predict how a given model will perform on a specific dataset. Our findings reveal several key principles:

- **The factorization component and node degree are Foundational.** Against the common assumption that the message-passing mechanism is the sole source of GNNs' power, our empirical results suggest that the factorization component (i.e., the quality of the node embeddings) and its most direct topological influence (**node degree**) are the most consistent and powerful predictors of performance. This implies that the primary benefit of GNNs often comes from learning robust node representations that are effectively regularized by their immediate neighborhood, a task for which even shallow models are well-suited.

- **Higher-order topologies explain performance differences.** While node degree sets the baseline, the ability to leverage more complex properties like **clustering coefficient** and **degree assortativity** is what often differentiates the performance of various GNN architectures. As our analysis revealed, models with more complex message-passing schemes are better equipped to implicitly capture these signals. The stronger the collaborative signal in a dataset (high clustering) or the more defined its homophily patterns (high assortativity), the greater the potential advantage of a GNN that can effectively explore multi-hop neighborhoods.

- **Use topology as a predictive tool to guide model selection.** The most practical application of this framework is as a diagnostic and predictive tool. Before launching extensive experiments, a topological analysis can guide resource allocation. For instance, if a dataset exhibits a low clustering coefficient and weak assortativity, it is highly unlikely that a complex, deep GNN will outperform a simpler, more efficient baseline. This allows practitioners to avoid wasting experimental cycles on architectures that are topologically mismatched with the data. The obtained results and explanatory

framework can help prevent situations where there is insufficient variability or richness in a topological property for a complex model to exploit, guiding the community toward more focused and justified model evaluation.